

Ajitkumar Senthil Kumar

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Full-stack & ML engineer with 2.4 years experience building production apps and RAG pipelines with React, FastAPI, PyTorch, and LangChain/Ollama.

Experience

Software Intern <i>Human CloudSoft, Maduravoyal, Chennai</i> - Improved inventory accuracy by 35% and cut audit time by 40% with automated alerts and streamlined workflows.. - Built a web-based inventory platform that transitioned all stock logs from manual to automated tracking.	May 2023 – Jun 2024
Frontend Developer <i>Klicknet Info Services Pvt Ltd, Korattur, Chennai</i> - Increased user retention by 25% by creating responsive, accessible UIs with React and TypeScript. - Improved cross-team efficiency by 30% by collaborating with backend teams to optimize workflows and performance.	Apr 2022 – Apr 2023
Design Engineer <i>L&T Rubber Processing Machinery, Vedal, Kanchipuram</i> - By converting complex 3D models into precise 2D AutoCAD drawings, I ensured production readiness and helped maintain defect-free outputs. - Reduced manufacturing errors by 15% and rework by 20% through detailed component modeling in CREO and the application of GD&T principles.	Nov 2021 – Feb 2022

Education

M.S., Data Science <i>State University of New York at Buffalo</i>	2024 – 2025 CGPA: 3.3/4
B.E., Mechanical Engineering <i>Sri Sairam Engineering College, Tambaram</i>	2017 – 2021 CGPA: 8.1/10

Projects

SmartLoanPred <i>Full-Stack Machine Learning App</i> - Developed a production-ready full-stack application for real-time loan approval prediction, using a React + Tailwind frontend and a FastAPI backend with secure JWT/OAuth authentication. - Deployed the Dockerized services via a GitHub Actions CI/CD pipeline with an Nginx reverse proxy.	Sep 2024 – Present
MIDAS <i>Full-Stack end-to-end ML Application</i> - Developed an end-to-end pipeline in PyTorch for dermatology image classification and risk screening, applying extensive image preprocessing, data augmentation with stratified sampling, and class-weighted loss to improve performance on a diverse dataset. - Architecting the transition from a research model to an ongoing product by implementing MLflow for scalable experiment tracking and model lifecycle management, while integrating Grad-CAM as a core feature to ensure interpretability and trust for medical professionals.	Jul 2024 – Aug 2024
CovDet <i>Computer Vision for Respiratory Disease</i> - Built a computer vision classifier to detect respiratory diseases from X-ray images, performing extensive image cleaning and augmentation while utilizing early stopping and cosine annealing to train robust, high-performance models. - Engineered a production-ready inference pipeline by exporting final models to efficient TorchScript/ONNX formats and deploying them via a FastAPI endpoint for real-time analysis.	May 2024 – Jun 2024
New York Taxi Rides <i>Predictive Modeling</i> - Modeled NYC taxi ride demand using historical trip data through extensive feature engineering with lag variables, rolling windows, and external joins (holidays, weather). - Used time-based cross-validation to ensure model robustness and integrated results into a Hopsworks feature store. - Developed a Streamlit dashboard for real-time monitoring and automated retraining workflows using GitHub Actions.	Feb 2024 – Apr 2024

Technical Skills

Programming: Python, R, SQL, JavaScript, TypeScript
Databases: SQL Server, PostgreSQL, Supabase, SQLite3
Frontend: React, Next.js, shadcn/ui, Tailwind CSS
Backend: FastAPI, SQLAlchemy
Machine Learning: Pandas, NumPy, scikit-learn, imbalanced-learn, PyTorch, Jupyter Notebook, RStudio
Cloud: AWS, DigitalOcean